Users' manual for DSurfTomo - 4D

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1, Description

DSurfTomo-4D is developed from the original DSurfTomo program for time-lapse surface-wave tomography. It uses two epochs of surface-wave dispersion data and directly inverts the shear-wave velocity change between them.
The original DSurfTomo program is used to obtain the reference velocity model at the first epoch, T1. DSurfTomo-4D then uses the T1 model together with the T1 and T2 dispersion data to recover the velocity perturbation.
For the data format, model format, grid definition, and general input parameters, please refer to the original DSurfTomo user manual.

2, Installation

Download the original DSurfTomo package from:
First, make a directory for the DSurfTomo code and example:
mkdir DSurfTomo
Second, clone the source code from Github with
cd DSurfTomo
git clone https://github.com/HongjianFang/DSurfTomo
At last, install
./configure
After successful compiling, the executable file will be in the 'bin' directory.

Download DSurfTomo-4D from:
git clone https://github.com/ZQ-de/DSurfTomo-4D.git
cd DSurfTomo-4D
./configure
After successful compilation, the DSurfTomo-4D executable is located in the `bin` directory.

3, Operation steps

Step 1: invert the first epoch
Use the original DSurfTomo program to invert the T1 dispersion data.
The required output is:
DSurfTomo.inMeasure.dat

Step 2: construct the reference model

Use the original initial `MOD` file as a template and replace its velocity values with those from:
DSurfTomo.inMeasure.dat
Save the updated model as:
MOD_T1
Before running DSurfTomo-4D, rename or copy it to:
MOD
For example:
cp MOD_T1 MOD

Step 3: prepare the two-epoch data
Save the T1 dispersion data as:
surfdata_ref.dat
Save the T2 dispersion data as:
surfdata_cur.dat
The two files should contain matched common source-receiver-period paths.

Step 4: prepare `DSurfTomo.in`
Keep the first data-file line as:
surfdata_ref.dat c: data file
Do not change this line. DSurfTomo-4D automatically reads:
surfdata_ref.dat
surfdata_cur.dat
The meanings of the other parameters are the same as in the original DSurfTomo program.

Step 5: run DSurfTomo-4D
Make sure the working directory contains:
DSurfTomo.in
MOD
surfdata_ref.dat
surfdata_cur.dat
Run:
./DSurfTomo-4D
The main output is:
DSurfTomo.deltaVs.dat
Its four columns are:
longitude latitude depth deltaVs
where `deltaVs` is the recovered shear-wave velocity change.